

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

Dec 2015

EC207 Control Systems

Date: 11.12.2015, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Discuss the difference between open loop control system and closed loop control system. [07]

Q - 2 (a) The characteristics equation of a system is given by [04]

$$3s^4 + 10s^3 + 5s^2 + 5s + 1 = 0$$

Determine the stability by means of Routh- Hurwitz criterion.

(b) Using Routh's criteria, for the unity feedback system, [05]

$$G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$$

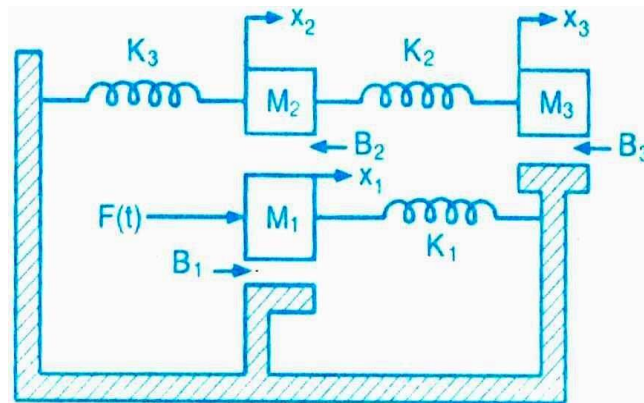
Find range of values of K.

(c) Find the stability of the system whose characteristics equation is [05]

$$s^5 + 2s^4 + 24s^3 + 48s^2 - 25s - 50 = 0$$

Also find the roots of this equation.

Q - 3 (a) For the following translation system shown in Figure, Obtain the differential equations and draw the electrical network using force-voltage analogy. [08]



(b) Define following with the help of neat diagram.

[06]

1. Phase cross-over
2. Phase cross-over frequency
3. Gain cross-over
4. Gain cross-over frequency
5. Gain Margin
6. Phase margin

OR

(b) Sketch the asymptotic bode plot for the transfer function given below and also find gain cross over frequency, phase cross over frequency, gain margin and phase margin of the system. Comment on Stability

$$G(s)H(s) = \frac{100}{s(s+1)(s+2)}$$

SECTION – II

Q-4 (a) Define the following terms.

[07]

- | | |
|-----------------------|------------------|
| 1) Transient Response | 5) Rise Time |
| 2) Damping Frequency | 6) Settling time |
| 3) Steady State Error | 7) Peak Time |
| 4) Transient Period | |

Q-5 (a) For unity feedback system having

[04]

$$G(s) = \frac{10(s+1)}{s^2(s+2)(s+10)}$$

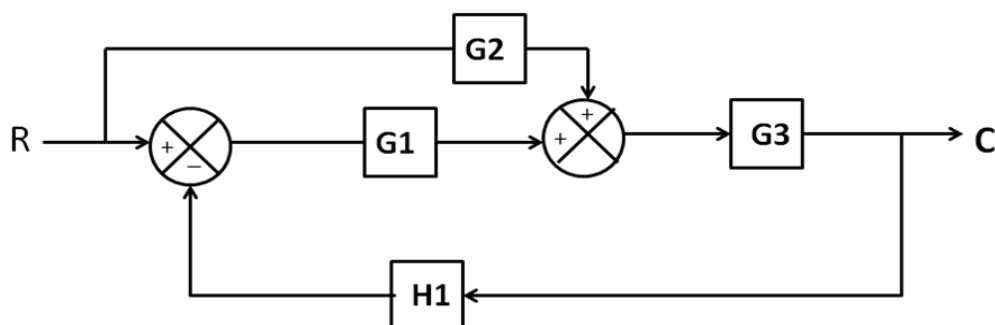
Determine :

- 1) Type of system
- 2) Error coefficients

(b) Answer any two questions.

[10]

(i) For the block diagram shown in figure below, determine the transfer function.



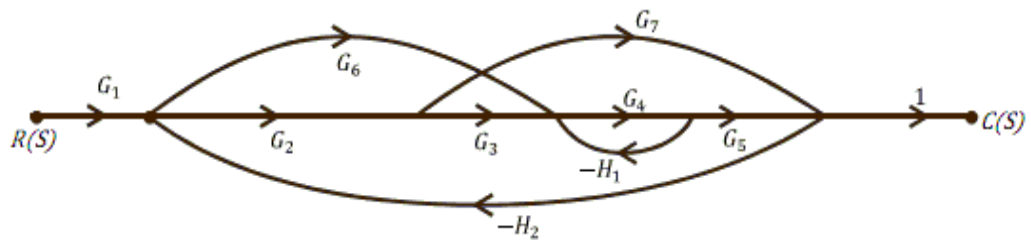
(ii) A system is described by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Test its observability.

(iii) Obtain transfer function from the signal flow graph shown in following figure (Use Mason's gain formula).



Q - 6 (a) Consider a unity feedback control system with an open loop transfer function [04]

$$G(s)H(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$$

Draw the Root Locus for the system with the gain K as variable.

(b) Answer any two questions. [10]

- (i) Give the differences between block diagram representation and signal flow graph representation of system.
- (ii) Define Peak overshoot and Derive its mathematical expression.
- (iii) Define Controllability and Observability of a system with appropriate example.
